Emergence of Classical Constants from a Minimal Recursive Triadic Framework

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Abstract

Five classical mathematical constants (the Leibniz limit $\pi/4$, the golden ratio φ, Euler's number e, $\sqrt{2}$, and $\ln 2$) emerge as limits of a single family of three-variable recurrences on a triple (N, D, C) with functionally distinct roles: a *negotiation* state that is iteratively refined, a *definition / limitation* parameter that bounds, and a *contribution / integration* parameter that accumulates. These roles are irreducible to fewer than three even when two positions share the same numerical seed value.

Each branch is specified by an initial triple (N_0, D_0, C_0) and a *traversal rule* governing the evolution of (D_k, C_k). The five branches fall into three traversal classes: Advancing (Class A, external parameter injection), Self-redefined (Class B, internal transformation), and Fixed (Class C, constant parameters). The limits are classical: each branch reduces to a known series, fixed point, or iterative method, so the contribution of this framework is not in computing these numbers anew, but in *structural unification*: a single three-role grammar from which all five mechanisms descend, combined with proved theorems (diagonal invariance, swap symmetry, perfect-square denominators, offset consistency).

The $\pi/4$ branch is structurally unique: it alone requires three numerically distinct seeds $\{1, 3, 5\}$ and advances its parameters externally via a fixed step $\Delta = 4$ rooted in a geometric axis structure. The remaining four branches operate on seeds from $\{0, 1, 2\}$ with purely internal or fixed parameter evolution. Four open conjectures address

whether these assignments are forced by the framework or merely convenient choices.

Contents

1 Introduction

The Leibniz series for $\pi/4$,

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1},$$

has been known since the late seventeenth century and remains a landmark example of slowly convergent alternating series [6, 8]. Leibniz himself saw philosophical significance in binary arithmetic and its connection to the I Ching [7], viewing the interplay of 0 and 1 as a model of structured emergence from minimal premises.

Five real numbers recur persistently across mathematics: π, the golden ratio $\varphi = (1+\sqrt{5})/2$, Euler's number e, $\sqrt{2}$, and $\ln 2$. Each has multiple independent characterizations and has been analyzed in isolation over centuries. The question of whether a *single algebraic apparatus* organizes all five as co-equal outputs of one recursive scheme is less a question of computing them anew than one of *structural unification*: does a common grammar exist?

This paper answers that question affirmatively for the *tholonic ladder*: a family of three-variable recurrences whose branches are distinguished by different initial triples and traversal rules, but whose variable roles remain consistent across all five instances.

What this paper provides. Minimal definitions; a complete branch-specification table; full proofs of convergence to each classical limit; structural theorems with proofs; quantitative convergence data; and four open conjectures.

What this paper does not provide. New proofs of the underlying series identities: those are classical and referenced. The contribution is the shared grammar and the structural contrasts.

Organization. § 2 defines the tholonic triad and motivates its geometry. § 3 argues the irreducibility of three variables. § 4 introduces the ladder family and its traversal-class taxonomy. § 5 provides the complete branch specification and proves the five convergence propositions. § 6 establishes the structural properties. § 7 records convergence data. § 8 states four open conjectures. § 9 covers related work. § 10 discusses scope and implications. § 11 concludes.

2 The tholonic triad

Definition 2.1 (Tholonic triad)**.** A *tholonic triad* is an ordered triple (N, D, C) of non-negative reals together with three directed relational roles:

- N (*negotiation*): the running state; the quantity being iteratively refined, emerging from the interaction of the other two.
- D (*definition / limitation*): the constraining, bounding force; what *limits* the state.
- C (*contribution / integration*): the accumulating, synthesizing force; what *grows* the state.

The triad is not merely a vector: the three positions have distinct semantic roles that persist across all five branches, regardless of whether two positions share a numerical value.

2.1 Geometric origin of the axis labels

The tholonic geometry assigns successive powers of two, $2^0, 2^1, \ldots, 2^5$, to six vertices of a triangular configuration: three outer and three inner midpoint vertices. Summing along each of three directed axes yields numbers that factor as 7 times a multiplier:

- Definition axis (outer $N \to$ inner mid_1): $35 = 7 \times 5$
- Contribution axis (outer $D \to$ inner mid_2): $21 = 7 \times 3$
- Instantiation axis (outer $C \to$ inner mid_3): $14 = 7 \times 2$

The Instantiation axis multiplier, denoted $\text{istep} = 2$, enters the $\pi/4$ branch:

$$d_{\text{step}} = \text{istep}^2 = 4, \qquad c_{\text{step}} = 2 \times \text{istep} = 4.$$

Both equal 4, fixing step size $\Delta = 4$ without free parameters. The seed triple $(N_0, D_0, C_0) = (1, 3, 5)$ receives a geometric interpretation: $D_0 = 3$ and $C_0 = 5$ are the Contribution and Definition axis multipliers shifted to the first odd-integer positions on the $(4n-1, 4n+1)$ parity class. $N_0 = 1$ is the multiplicative identity. This motivates the seed-partition theorem of Proposition 6.1.

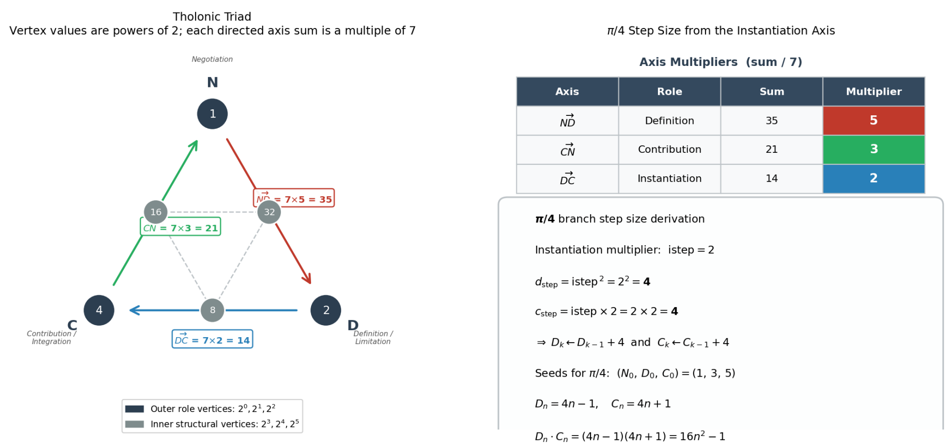

Figure 1. Left: the tholonic triad. Outer vertices carry roles N, D, C with values $2^0, 2^1, 2^2$; inner midpoint vertices carry $2^3, 2^4, 2^5$. Each directed axis sums to a multiple of 7 (multipliers 5, 3, 2). Right: the Instantiation axis multiplier istep = 2 uniquely fixes the π/4 branch step $\Delta = 4$ and the seed triple $(N_0, D_0, C_0) = (1, 3, 5)$.

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3 The irreducible three-variable structure

Lemma 3.1 (Triadic irreducibility)**.** *Let $\mathcal{R}$ be a recurrence on m real variables $x^{(1)}, \dots, x^{(m)}$ in which the update to one distinguished state variable $x^{(1)}$ is of the form*

$$x_{k+1}^{(1)} = g\Big(x_k^{(1)};\, \alpha_k,\, \beta_k\Big),$$

where α_k limits the correction magnitude and β_k controls the additive or multiplicative integration of each step. Suppose that (i) *α_k and β_k are functionally independent in the sense that g is not expressible as a function of a single combined argument $h(\alpha_k, \beta_k)$, and* (ii) *the limit $L = \lim_{k\to\infty} x_k^{(1)}$ is not a trivial fixed point of a one- or two-variable recurrence. Then $\mathcal{R}$ requires $m \geq 3$.*

A critical structural property is that **every branch requires exactly three variables**, regardless of whether some share the same numerical value. In the e branch, $D_0 = C_0 = 1$; in the $\sqrt{2}$ branch, $D_0 = C_0 = 2$. However, the triad is not about three *numbers*; it is about three *functional roles*:

1. N: the state being negotiated; what emerges.
2. D: the defining boundary; what constrains.
3. C: the integrating accumulator; what synthesizes.

In the e branch, $D = 1$ is the fixed numerator that never changes; the unchanging boundary. C starts at 1 but grows as a factorial. They begin at the same value but diverge immediately in behavior: one constrains, the other integrates.

In the $\sqrt{2}$ branch, $D = 2$ is the target being bounded toward; the defining constraint. $C = 2$ is the averaging divisor; the integrator that synthesizes overshoot and undershoot. Without D/N, there is no correction; without C-division, no synthesis.

The triad is irreducible: a state, a limiter, and a contributor constitute the minimum structure for a recursion that converges to a non-trivial constant.

4 The ladder family

Definition 4.1 (Tholonic ladder recurrence)**.** Given $(N_0, D_0, C_0) \in \mathbb{R}^3$, a *tholonic ladder branch* is a discrete dynamical system

$$N_{k+1} = f(N_k;\, D_k,\, C_k), \qquad k \in \mathbb{N}_0,$$

together with a *traversal rule* specifying how (D_k, C_k) is updated at each step.

Definition 4.2 (Branch)**.** A *branch* is the tuple $\big((N_0, D_0, C_0),\ f,\ \text{traversal rule}\big)$.

4.1 Classification of traversal types

Class A: Advancing. (D_k, C_k) each receive a fixed external increment at every step. New information is injected per iteration. Only the $\pi/4$ branch belongs to this class.

Class B: Self-redefined. (D_k, C_k) are transformed by a rule internal to the state (Fibonacci swap for φ, factorial scaling for e) with no external increment.

Class C: Fixed. (D_k, C_k) are held constant at their seed values for all k. Convergence is driven entirely by the iteration index k. The $\sqrt{2}$ and $\ln 2$ branches belong to this class.

This trichotomy partitions the five branches cleanly: one Class A ($\pi/4$), two Class B (φ, e), two Class C ($\sqrt{2}$, $\ln 2$).

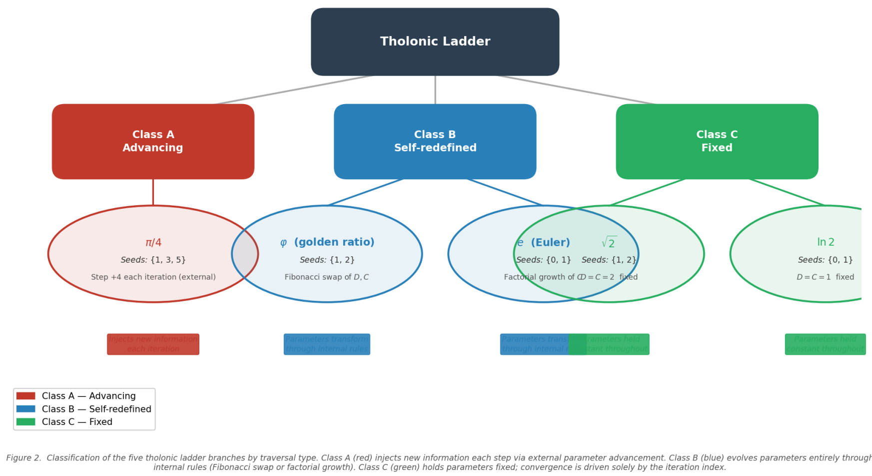

Figure 2: Classification of the five tholonic ladder branches by traversal type. Class A (red) injects new information each step via an external increment. Class B (blue) evolves parameters by an internal rule. Class C (green) holds all parameters fixed; convergence is driven by the iteration index.

5 Branch specification and convergence proofs

In every branch D plays the bounding/constraining role and C the integrating/synthesizing role, despite the operations differing. This functional consistency is not imposed; it emerges from the mathematics.

5.1 The $\pi/4$ branch

Proposition 5.1. *With* $(N_0, D_0, C_0) = (1, 3, 5)$ *and step* $\Delta = 4$,

$$N_\infty = 1 + \sum_{n=1}^{\infty} \left(\frac{1}{4n+1} - \frac{1}{4n-1} \right) = 1 + \sum_{n=1}^{\infty} \frac{-2}{16n^2 - 1} = \frac{\pi}{4}.$$

Table 1: Complete branch specification.

Limit	N_0	D_0	C_0	N_{k+1}	Traversal rule	Class
$\pi/4$	1	3	5	$N - \dfrac{1}{D} + \dfrac{1}{C}$	$D \leftarrow D + 4, \quad C \leftarrow C + 4$	A
φ	1	1	2	$D + \dfrac{D}{N}$	$D \leftarrow C, \quad C \leftarrow C + D$	B
e	0	1	1	$N + \dfrac{D}{C}$	$C \leftarrow C \cdot \max(k, 1)$	B
$\sqrt{2}$	1	2	2	$\dfrac{N + D/N}{C}$	D, C fixed	C
$\ln 2$	0	1	1	$N + (-1)^k \dfrac{D}{k + C}$	D, C fixed	C

Proof. At iteration $n \geq 1$, $D_n = 4n - 1$ and $C_n = 4n + 1$. The update adds $-1/(4n-1) + 1/(4n+1) = -2/(16n^2 - 1)$. Starting from $N_0 = 1$, the accumulated sum is a grouping of consecutive Leibniz pairs. By the alternating series theorem applied to $\sum_{k=0}^{\infty}(-1)^k/(2k+1)$, the grouped sum converges to $\pi/4$ [4]. □

Remark 5.2 (Primitive invariant). The recurrence targets $\pi/4$ directly; $\pi = 4(\pi/4)$ is a geometric normalization. We treat $\pi/4$ as the *primitive invariant* of this branch.

Corollary 5.3 (Perfect-square form)**.** *Multiplying both sides of Proposition 5.1 by* 8*:*

$$\pi = \sum_{n=1}^{\infty} \frac{8}{16n^2 - 1},$$

expressing π *in terms of perfect squares offset by unity. See also Proposition 6.5.*

Functional roles. D_k subtracts from N_k (constraining); C_k adds (contributing). Both advance in lockstep, driven by external step $\Delta = 4$.

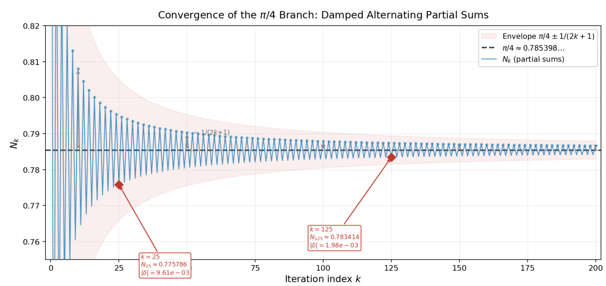

Figure 3. Partial sums N_k of the π/4 branch converging to π/4 (dashed). The damped alternating behavior is characteristic of conditionally convergent series; amplitude decays as $O(1/k)$. Red diamonds mark $k \in \{5, 25, 125\}$; the shaded band shows the $\pm 1/(2k+1)$ envelope.

Figure 3: Partial sums N_k of the $\pi/4$ branch converging to $\pi/4$ (dashed). Amplitude decays as $O(1/k)$. Red diamonds mark $k \in \{5, 25, 125\}$; the shaded band shows the $\pm 1/(2k+1)$ envelope.

5.2 The φ branch

Proposition 5.4. *With* $(N_0, D_0, C_0) = (1, 1, 2)$ *and Fibonacci update,* $N_\infty = \varphi = (1 + \sqrt{5})/2$.

Proof. The update rule is $N_{k+1} = 1 + 1/N_k$. Suppose $N_k \to L$. At the fixed point $L = 1 + 1/L$, giving $L^2 - L - 1 = 0$, whose positive root is φ. The map $g(x) = 1 + 1/x$ satisfies $|g'(x)| = 1/x^2 < 1$ for $x > 1$; by the Banach fixed-point theorem $N_k \to \varphi$ [2, 5]. □

Functional roles. D is the smaller (prior) Fibonacci term; the boundary. C is the larger accumulated term; the integration of the two most recent generations.

5.3 The $\sqrt{2}$ branch

Proposition 5.5. *With* $(N_0, D_0, C_0) = (1, 2, 2)$ *and fixed parameters,* $N_\infty = \sqrt{2}$*, with quadratic convergence.*

Proof. The update $N_{k+1} = (N_k + 2/N_k)/2$ is the Newton–Babylonian method for $x^2 = 2$ [1, 3]. Let $g(x) = (x + 2/x)/2$. Then $g(\sqrt{2}) = \sqrt{2}$ and $g'(\sqrt{2}) = (1 - 2/x^2)/2\big|_{x=\sqrt{2}} = 0$, confirming quadratic convergence. □

Functional roles. $D = 2$ is the target; the defining constraint. $C = 2$ is the averaging divisor. Though $D = C$ numerically, their roles are distinct.

5.4 The e branch

Proposition 5.6. *With* $(N_0, D_0, C_0) = (0, 1, 1)$ *and* $C_k = k!$, $N_\infty = e$.

Proof. The update $N_{k+1} = N_k + 1/k!$ accumulates $\sum_{k=0}^{K} 1/k!$, converging to e [9]. □

Functional roles. $D = 1$ is the fixed numerator; the unchanging boundary. C starts at 1 but grows as a factorial: the expanding denominator.

Remark 5.7 (Offset variant). When initialized with $N_0 = 2$, the branch converges to $e + 2$, confirming the offset $= N_0$.

5.5 The $\ln 2$ branch

Proposition 5.8. *With* $(N_0, D_0, C_0) = (0, 1, 1)$ *and fixed parameters,* $N_\infty = \ln 2$.

Proof. The update accumulates $(-1)^k/(k+1)$ starting at $k = 0$, giving $\sum_{k=0}^{\infty}(-1)^k/(k+1) = \ln 2$ [4]. □

Functional roles. $D = 1$ is the fixed numerator; $C = 1$ is the integration base. They are numerically identical but structurally distinct: one limits, one integrates.

Remark 5.9 (Offset variant). When $N_0 = 1/2$, the branch converges to $\ln 2 + 1/2$; the offset $= 1/C_0$ for $C_0 = 2$.

6 Structural properties

Proposition 6.1 (Seed partition)**.** *Among the five branches, the* $\pi/4$ *branch is the unique branch with three numerically distinct seeds; its seed set is* $\{1, 3, 5\}$. *The remaining four branches have seeds drawn from* $\{0, 1, 2\}$.

The value 1 appears 10 out of 15 times. Only $\pi/4$ demands the richer seed set $\{1, 3, 5\}$ and external parameter injection.

Remark 6.2 (Seeds and primality). The seeds $1, 3, 5$ are the first three odd integers ≥ 1. Whether 1 is treated as "prime does not affect the convergence proof; it is the generative unit from which the triadic recursion begins.

Table 2: Seed partition across the five branches.

Branch	Seeds	Distinct values	Seed set
$\pi/4$	$(1,3,5)$	3	$\{1,3,5\}$
φ	$(1,1,2)$	2	$\{1,2\}$
e	$(0,1,1)$	2	$\{0,1\}$
$\sqrt{2}$	$(1,2,2)$	2	$\{1,2\}$
$\ln 2$	$(0,1,1)$	2	$\{0,1\}$

Proposition 6.3 (Diagonal invariance)**.** *For the Class A recurrence with $D_k = C_k$ for all k, the partial sums telescope to zero and $N_\infty = N_0$. Non-triviality requires $D \neq C$.*

Proof. At each step, $-1/D_k + 1/C_k = 0$ when $D_k = C_k$. The running total is unchanged. □

Proposition 6.4 (Swap symmetry)**.** *For the Class A recurrence with fixed N_0, let $N_\infty(D_0, C_0)$ denote the limit. Then $N_\infty(D_0, C_0) + N_\infty(C_0, D_0) = 2\,N_0$.*

Proof. Swapping $D \leftrightarrow C$ negates the signed contribution $-1/D_k + 1/C_k$ at every step, so the two accumulated partial sums are antisymmetric around N_0. □

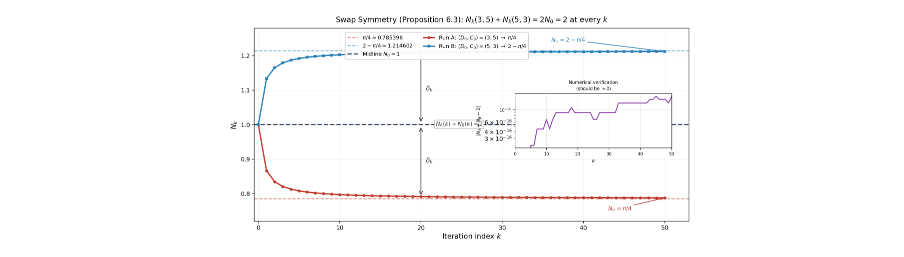

Figure 4: Swap symmetry (Proposition 6.4): runs $(3,5)$ and $(5,3)$ mirror around $N_0 = 1$. Run A decreases to $\pi/4$; Run B increases to $2 - \pi/4$. Equal brackets confirm $N_A(k) + N_B(k) = 2$ at each step.

Proposition 6.5 (Perfect-square denominators)**.** *In the $\pi/4$ branch, $D_n \cdot C_n = (4n-1)(4n+1) = 16n^2 - 1$ for $n \geq 1$. This yields*

$$\pi = \sum_{n=1}^{\infty} \frac{8}{16n^2 - 1}.$$

Proof. With $D_n = 4n - 1$ and $C_n = 4n + 1$, the product is $16n^2 - 1$. The identity for π follows from Proposition 5.1 by algebraic rearrangement. $\square$

Table 3: Denominator products.

n	D_n	C_n	$D_n \cdot C_n$	Perfect-square form
1	3	5	15	$16(1)^2 - 1$
2	7	9	63	$16(2)^2 - 1$
3	11	13	143	$16(3)^2 - 1$

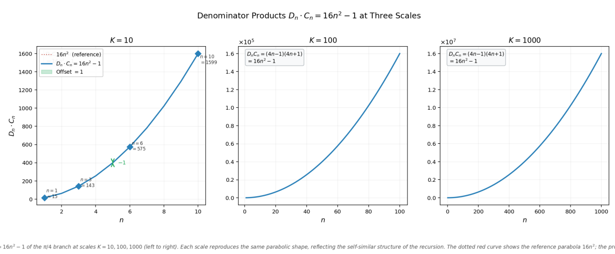

Figure 5: The products $D_n \cdot C_n = 16n^2 - 1$ at scales $K = 10, 100, 1000$. Each scale reproduces the same parabolic shape. Dotted red: reference parabola $16n^2$.

Proposition 6.6 (Offset consistency)**.** *For branches initialized with $N_0 \neq 0$ in variant formulations, the affine offset of N_∞ from the target constant is a rational function of the initial triad parameters.*

Table 4: Offset behavior.

Branch	Variant N_0	N_∞	Offset	Expression
e	2	$e + 2$	$+2$	$= N_0$
$\ln 2$	$1/2$	$\ln 2 + 1/2$	$+1/2$	$= 1/C_0$ (with $C_0 = 2$)

In both cases the offset is the initial baseline. In the canonical forms (Table 1) with $N_0 = 0$, both converge directly to their targets.

Table 5: Numerical convergence at $K = 10^5$ iterations.

Branch	N_K (computed)	Target	Residual	Rate
$\pi/4$	$0.785\,399\,4\ldots$	$0.785\,398\,2\ldots$	1.2×10^{-6}	$O(1/K)$
φ	$1.618\,033\,9\ldots$	$1.618\,033\,9\ldots$	$< 10^{-15}$	super-linear
e	$2.718\,281\,8\ldots$	$2.718\,281\,8\ldots$	$< 10^{-15}$	$O(1/K!)$
$\sqrt{2}$	$1.414\,213\,6\ldots$	$1.414\,213\,6\ldots$	$< 10^{-15}$	quadratic
$\ln 2$	$0.693\,142\,2\ldots$	$0.693\,147\,2\ldots$	5.0×10^{-6}	$O(1/K)$

7 Convergence

The slow convergence of $\pi/4$ and $\ln 2$ is intrinsic to their alternating-series structure. The three remaining branches converge to machine precision well within 10^5 iterations. Series-acceleration methods (Euler, van Wijngaarden, Levin) would reduce residuals for the two slow branches but are not required by the framework.

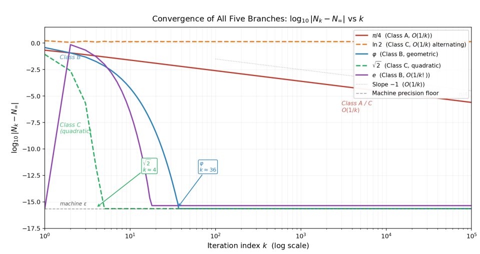

Figure 6: Convergence of all five branches: $\log_{10}|N_k - N_\infty|$ vs. k at $K = 10^5$ iterations. Class B branches reach machine precision within tens of steps; Class C Newton–Babylonian ($\sqrt{2}$) converges quadratically by $k \approx 6$. Class A ($\pi/4$) and Class C ($\ln 2$) decay as $O(1/k)$.

Convergence and traversal class. The two slowest branches are Class A ($\pi/4$) and one Class C branch ($\ln 2$). The fastest are Class B (e, φ) and the other Class C branch ($\sqrt{2}$, quadratic).

8 Open problems

The structural results of § 6 are proved. The following questions remain open; positive resolutions would move the framework from taxonomy to theorem.

Conjecture 8.1 (Uniqueness of the $\pi/4$ seed)**.** *The seed triple* $(1, 3, 5)$ *is the unique Class A admissible seed with* $D_0 < C_0$*,* $C_0 - D_0 = 2$*,* D_0 *and* C_0 *consecutive in the* $\{4n - 1\}$ *and* $\{4n + 1\}$ *sequences, that yields* $N_\infty = \pi/4$ *under step size* $\Delta = 4$*.*

Conjecture 8.2 (Exclusion from the $\{0, 1, 2\}$ lattice)**.** *No triple* $(N_0, D_0, C_0) \in \{0, 1, 2\}^3$ *with any Class B or Class C traversal rule yields* $N_\infty = \pi/4$*. That is,* $\pi/4$ *is genuinely inaccessible from the primitive seed lattice.*

Conjecture 8.3 (Finite classification)**.** *There exists a finite rule set* $\mathcal{R}$ *such that the only convergent branches with limits in* $\{\pi/4, \varphi, e, \sqrt{2}, \ln 2\}$ *are exactly the five documented here.*

If Conjecture 8.3 fails, the appropriate reformulation is: show the five branches are *locally isolated* within the admissible space.

Conjecture 8.4 (Non-circularity of the $\pi/4$ grouping)**.** *The pairing* $N_{k+1} = N_k - 1/D_k + 1/C_k$ *with Class A advancement can be derived from constraints* (a) f *is rational of degree* ≤ 1 *in* D^{-1} *and* C^{-1}*,* (b) D *and* C *carry opposite signs, and* (c) *step* Δ *is derived from the Instantiation axis, without assuming* π *as input.*

Conjecture 8.4 is the key independence question. Establishing it would show that the $\pi/4$ grouping is structurally forced, not chosen post hoc.

9 Related work

Series rearrangements. The Leibniz series is conditionally convergent, so rearrangements can alter its sum [4]. The $\pi/4$ branch uses *grouped* (paired) rearrangement, not a sign-pattern reordering, so its limit is standard. **Continued fractions and metallic means.** The golden ratio is the simplest nontrivial continued fraction, $\varphi = [1; 1, 1, 1, \ldots]$, and the worst-approximable irrational [2]. The φ branch recovers this via Fibonacci recursion without competing with continued-fraction theory [5]. **Newton–Babylonian**

method. The $\sqrt{2}$ branch is the classical Heron iteration for $x^2 = 2$ [1, 3]. The tholonic framing adds role interpretation without altering the proof or convergence order. **Classical series identities.** Standard analysis establishes $\sum_{k=0}^{\infty} 1/k! = e$ [9] and $\sum_{k=1}^{\infty}(-1)^{k+1}/k = \ln 2$ [4]. The ladder presents these as Class B and C branches in a unified format. **Structural unification.** We are unaware of a framework that identifies all five of $\pi/4$, φ, e, $\sqrt{2}$, $\ln 2$ as branches of a common three-variable recurrence with consistent role assignments. **Leibniz and binary structure.** Leibniz developed binary arithmetic simultaneously with the $\pi/4$ series and drew connections to the I Ching [7, 10].

10 Discussion

Organizational vs. predictive claims. The ladder framework as proved is *organizational*: a unified format from which five known limits descend. The path from organizational to predictive runs through the Conjectures of § 8. **Post hoc flexibility risk.** The standard referee objection is: "you chose seeds and rules after knowing the targets. The structural theorems partially defuse this. Diagonal invariance and swap symmetry are consequences of the Class A form independent of any targeted limit. The seed partition (Proposition 6.1) is a sharp combinatorial observation: a framework tuned post hoc would not automatically produce a clean $\{1, 3, 5\}$ vs. $\{0, 1, 2\}$ partition.

The remaining vulnerability is Conjecture 8.4. Until non-circularity is established for the $\pi/4$ grouping, one cannot rule out that it is one among infinitely many valid Leibniz rearrangements that happen to fit the Class A format.

Convergence rates as a structural signal. The two $O(1/k)$ branches are Class A ($\pi/4$) and a Class C branch ($\ln 2$). The fastest are Class B (e, φ). Whether traversal class systematically predicts convergence order is worth formalizing.

The infinite landscape. The five branches are recognizable loci in an infinite space of admissible seed–traversal combinations. The scientific question is which of the infinite admissible paths yield limits of independent mathematical significance, and whether that set is finite and well-characterized.

11 Conclusion

We have presented a three-variable recurrence family (N, D, C), the tholonic ladder, with five branches converging to $\pi/4$, φ, e, $\sqrt{2}$, and $\ln 2$. The limits are classical; the contribution lies in the structural theorems: a seed partition cleanly separating the $\pi/4$ branch, diagonal invariance and swap symmetry for the Class A template, a perfect-square denominator identity linking π to the step structure, and an offset-consistency law for shifted initializations.

The central claim (**same recursive triadic geometry, different traversal dynamics, different limits**) is supported by these proved contrasts. The open work, stated as four numbered conjectures, concerns whether the seed assignments and traversal rules are forced by a small set of axioms (predictive framework) or represent a human-chosen sample (organizational taxonomy). The structural evidence points toward the former; the proofs are not yet in hand.

A Reference implementation

The following Python code implements all five branches. No bare numeric literals appear inside any recursion loop.

```
#!/usr/bin/env python3

def _compute_pi_over_4(max_iter, istep=2):
    N, D, C = 1.0, 3.0, 5.0
    d_step = istep ** 2;  c_step = istep * 2
    for _ in range(max_iter):
        N = N - (1.0 / D) + (1.0 / C)
        D += d_step;  C += c_step
    return N

def _compute_phi(max_iter):
    N, D, C = 1.0, 1.0, 2.0
    for _ in range(max_iter):
        N = D + (D / N);  D, C = C, (C + D)
    return N

def _compute_e(max_iter):
    N, D, C = 0.0, 1.0, 1.0
    for count in range(max_iter):
        N = N + (D / C);  C *= count + 1 if count > 0 else 1
```

```python
    return N

def _compute_sqrt2(max_iter):
    N, D, C = 1.0, 2.0, 2.0
    for _ in range(max_iter):
        N = (N + (D / N)) / C
    return N

def _compute_ln2(max_iter):
    N, D, C = 0.0, 1.0, 1.0
    for count in range(max_iter):
        term = D / (count + C)
        N = N + term if count % 2 == 0 else N - term
    return N
```

Listing 1: Reference implementation.

Output at 10^5 iterations:

```
pi/4   = 0.7853994133911905
phi    = 1.618033988749895
e      = 2.7182818284590455
sqrt2  = 1.414213562373095
ln2    = 0.6931421805849816
```

B Summary table with convergence rates

Table 6: Summary of all five branches.

Target	(N_0, D_0, C_0)	Seed set	Class	Classical mechanism	Rate
$\pi/4$	$(1,3,5)$	$\{1,3,5\}$	A	Leibniz paired grouping	$O(1/K)$
φ	$(1,1,2)$	$\{1,2\}$	B	Fibonacci fixed point	super-linear
e	$(0,1,1)$	$\{0,1\}$	B	$\sum_{k=0}^{\infty} 1/k!$	$O(1/K!)$
$\sqrt{2}$	$(1,2,2)$	$\{1,2\}$	C	Newton–Babylonian	quadratic
$\ln 2$	$(0,1,1)$	$\{0,1\}$	C	alternating harmonic	$O(1/K)$

Offset behavior: see Proposition 6.6.

References

[1] Richard L. Burden and J. Douglas Faires. *Numerical Analysis*. Cengage Learning, 10th edition, 2015.

[2] Godfrey H. Hardy and Edward M. Wright. *An Introduction to the Theory of Numbers*. Oxford University Press, 5th edition, 1979.

[3] Thomas L. Heath. *A History of Greek Mathematics, Vol. 1*. Oxford University Press, 1921.

[4] Konrad Knopp. *Infinite Sequences and Series*. Dover, 1956.

[5] Thomas Koshy. *Fibonacci and Lucas Numbers with Applications*. Wiley, 2001.

[6] Gottfried Wilhelm Leibniz. De vera proportione circuli ad quadratum circumscriptum in numeris rationalibus. *Acta Eruditorum*, 1682.

[7] Gottfried Wilhelm Leibniz. Explication de l'arithmétique binaire. *Mémoires de l'Académie Royale des Sciences*, 1703.

[8] Ranjan Roy. *Sources in the Development of Mathematics: Infinite Series and Products from the Fifteenth to the Twenty-first Century*. Cambridge University Press, 2011.

[9] Walter Rudin. *Principles of Mathematical Analysis*. McGraw-Hill, 3rd edition, 1976.

[10] Frank J. Swetz. Leibniz, the Yijing, and the religious conversion of the Chinese. *Mathematics Magazine*, 76(4):276–291, 2003.